MATH 174 - Numerical Analysis

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Contents

1 Error

1.1 Approximation

Let x be an exact number.
Let y be an approximation of x.

Def: The absolute error is $|x - y|$.
Def: The relative error is $\dfrac{|x-y|}{|x|}$,

1.2 Round-off Error

Usually a machine uses a finite amount of memory to store a number.
machine $\leftarrow \pi$
Suppose a machine can store some number of digits.

1. Allocate a digit for sign, positive or negative.

2. Allocate a number of digits for the mantissa.
3. Allocate a digit for sign of the exponent.
4. Allocate a number of digits for the exponent.

Under rounding in the case of a 4 digit mantissa, $\pi = 0.3142 \cdot 10^1$, this is the machine # representation of pi.

Def: Round-off error are all errors arising from the error between actual # and machine #.

Notation:

$$x = \text{real } \#$$
$$fl(x) = \text{machine } \# \text{ representation of } x$$

In an s-digit rounding machine,
Abs error $= |x = fl(x)|$
Rel error $= \dfrac{|x - fl(x)|}{|x|} \leq \frac{1}{2} \cdot 10^{1-5} = 5 \cdot 10^{-5}$

1.3 Floating Point Arithmetic

What happens under arithmetic operations? (+, -, ·, /)

Two things to watch out for:

1. Many operations can degrade error.
2. Catastrophic cancellation.

1.3.1 Degrading Error

number	machine
x	$fl(x)$
y	$fl(y)$
$x + y$	$fl(fl(x) + fl(y))$

The more nested $fl()$ there are, the larger the error will be. The same follows for -, ·, /.

When performing arithmetic, it temporarily allows for more digits than the mantissa allows.

Algorithmic "solutions"

$((x + y) + z) + w$

Worst case, 4 times the error.

$(x + y) + (z + w)$

Worst case, 3 times the error.

1.3.2 Catastrophic Cancellation

Ex: 3-digit rounding machine. Given 2 numbers: 0.312 and 0.311.

$$0.312 - 0.311 = 0.001 = 0.1 \cdot 10^{-2}$$

Loss of significant digits!

Ex: 5-digit rounding machine. Given 2 numbers:

$$\begin{aligned}
x &= 0.41626324 \\
y &= 0.41626323 \\
x - y &= 0.00000001 \\
\text{In machine:} & \\
fl(x) &= 0.41626 \\
fl(y) &= 0.41626 \\
fl(x) - fl(y) &= 0.00000 = 0 \\
fl(fl(x) - fl(y)) &= 0 \\
\text{Relative error:} & \\
\frac{|x - y - fl(fl(x) - fl(y))|}{|x - y|} &= \frac{|x - y - 0|}{|x - y|} \\
&= 1 = 100\% \text{ error!}
\end{aligned}$$